COMPASS_TEMPEST_SGW_2023: April

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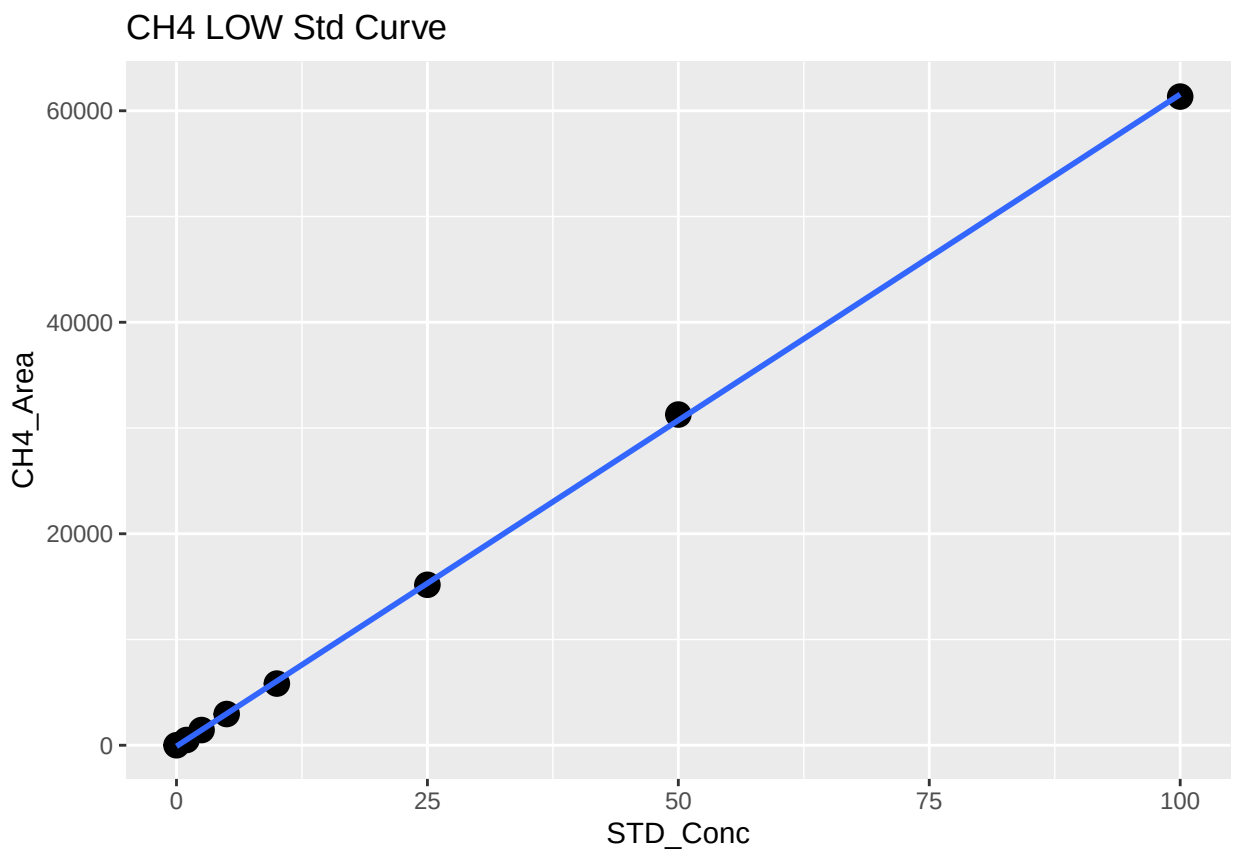
2023-01-14

##Set Up

Read in first data file and assess standard curves

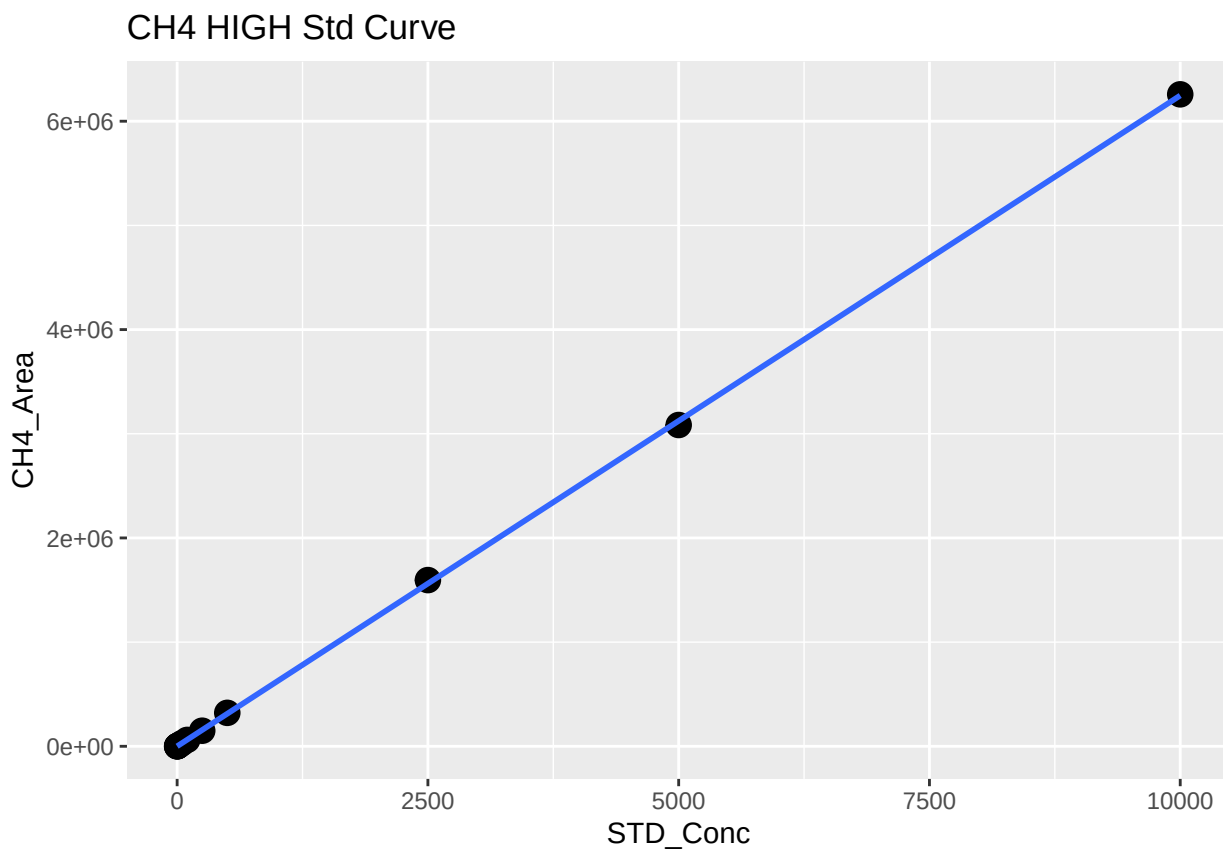
```
##     Machine       User Run_Date Sample_Type Sample1 Sample_Year Sample_Month
## 1 Varian GC Max Wegner 20231121       Blank   Blank        2023         <NA>
## 2 Varian GC Max Wegner 20231121     STD_CH4 STD_CH4        2023         <NA>
## 3 Varian GC Max Wegner 20231121     STD_CO2 STD_CO2        2023         <NA>
## 4 Varian GC Max Wegner 20231121     STD_CH4 STD_CH4        2023         <NA>
## 5 Varian GC Max Wegner 20231121     STD_CH4 STD_CH4        2023         <NA>
## 6 Varian GC Max Wegner 20231121     STD_CH4 STD_CH4        2023         <NA>
##   Sample_Day Sample_Time Time_Zone G_W Sample_Plot Sample_Grid Sample_Depth
## 1         NA          NA      <NA>   G        <NA>        <NA>         <NA>
## 2         NA          NA      <NA>   G        <NA>        <NA>         <NA>
## 3         NA          NA      <NA>   G        <NA>        <NA>         <NA>
## 4         NA          NA      <NA>   G        <NA>        <NA>         <NA>
## 5         NA          NA      <NA>   G        <NA>        <NA>         <NA>
## 6         NA          NA      <NA>   G        <NA>        <NA>         <NA>
##              Sample_ID Dilution_Factor STD_Conc CO2_Area CH4_Area Notes
## 1              Blank_0               1      0.0        0        0    NA
## 2 Blank_0_repeatforCH4               1      0.0        0        0    NA
## 3 Blank_0_repeatforCO2               1      0.0        0        0    NA
## 4         STD_1ppm_CH4               1      1.0    21878      497    NA
## 5       STD_2.5ppm_CH4               1      2.5    64189     1439    NA
## 6         STD_5ppm_CH4               1      5.0   135079     2952    NA
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



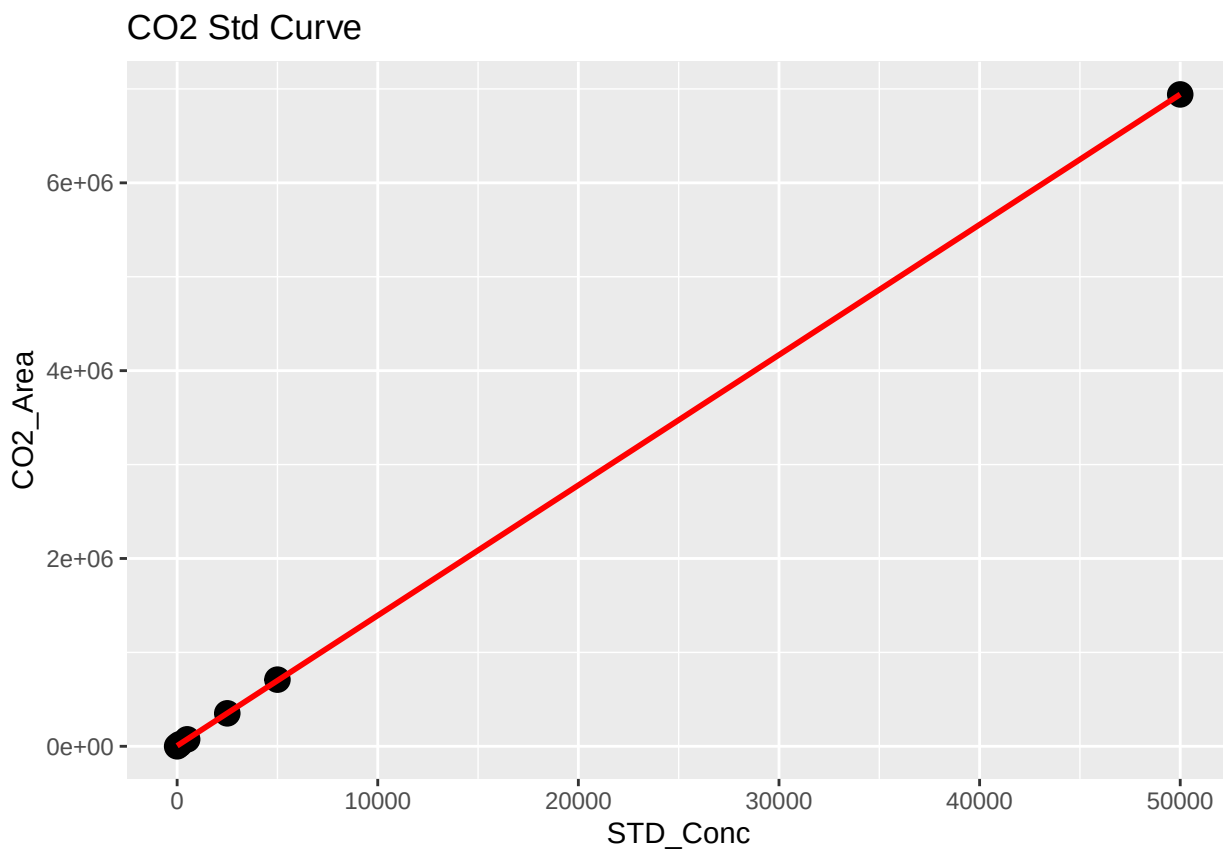

```
##
## Call:
## lm(formula = stds_ch4_low$CH4_Area ~ stds_ch4_low$STD_Conc)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -235.49 -158.31  -26.35   21.99  545.69
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)            -98.713    117.564   -0.84    0.433
## stds_ch4_low$STD_Conc  616.520      2.888  213.48 7.13e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 267.5 on 6 degrees of freedom
## Multiple R-squared:  0.9999, Adjusted R-squared:  0.9998
## F-statistic: 4.557e+04 on 1 and 6 DF,  p-value: 7.129e-13
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

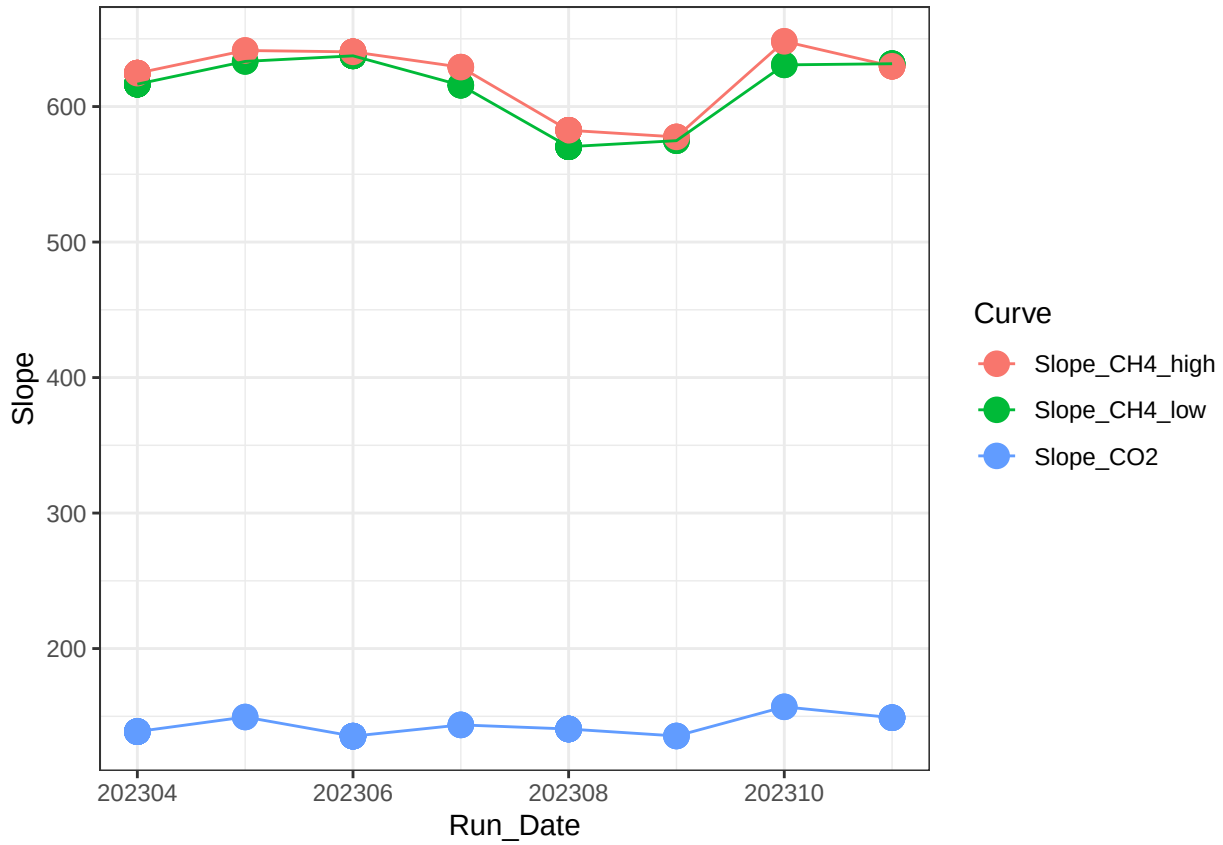



```
##
## Call:
## lm(formula = stds_ch4$CH4_Area ~ stds_ch4$STD_Conc)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -39658   -930   -616   -447  33345
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)         488.163   5061.596   0.096    0.925
## stds_ch4$STD_Conc   624.638      1.591 392.607   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 16330 on 11 degrees of freedom
## Multiple R-squared:  0.9999, Adjusted R-squared:  0.9999
## F-statistic: 1.541e+05 on 1 and 11 DF,  p-value: < 2.2e-16
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



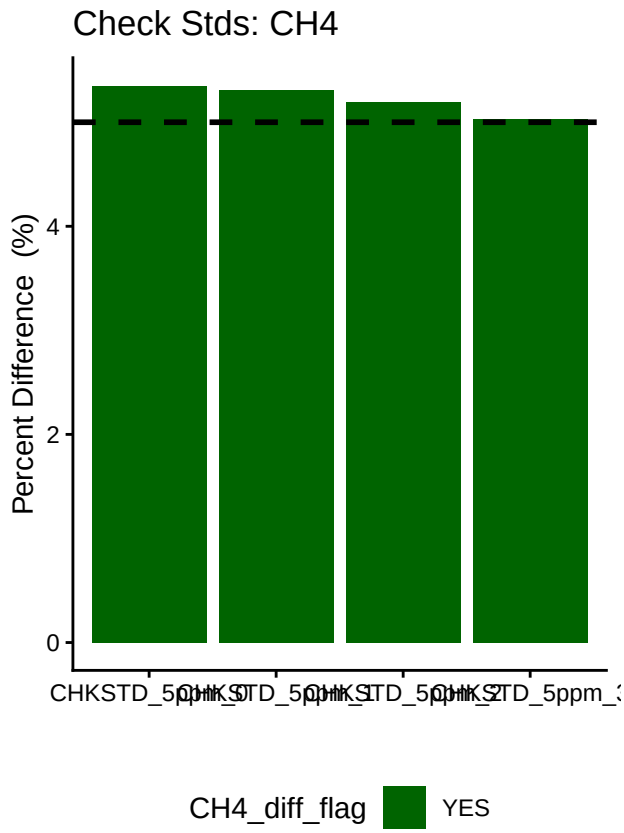

```
##
## Call:
## lm(formula = stds_co2$CO2_Area ~ stds_co2$STD_Conc)
##
## Residuals:
##       1       2       3       4       5       6
## -5556.9 -1700.2  -864.1 -1410.8 10498.3  -966.4
##
## Coefficients:
##                     Estimate Std. Error  t value Pr(>|t|)
## (Intercept)       5556.8684  2812.8555    1.976    0.119
## stds_co2$STD_Conc  138.7384     0.1369 1013.127 5.69e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6076 on 4 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:      1
## F-statistic: 1.026e+06 on 1 and 4 DF,  p-value: 5.695e-12

##   X          Curve        R2    Slope  Intercept Run_Date
## 1 1  Slope_CH4_low 0.9998464 616.5204  -98.71283   202304
## 2 2 Slope_CH4_high 0.9999222 624.6385  488.16259   202304
## 3 3      Slope_CO2 0.9999951 138.7384 5556.86838   202304
## 4 4  Slope_CH4_low 0.9998464 616.5204  -98.71283   202304
## 5 5 Slope_CH4_high 0.9999222 624.6385  488.16259   202304
## 6 6      Slope_CO2 0.9999951 138.7384 5556.86838   202304
```

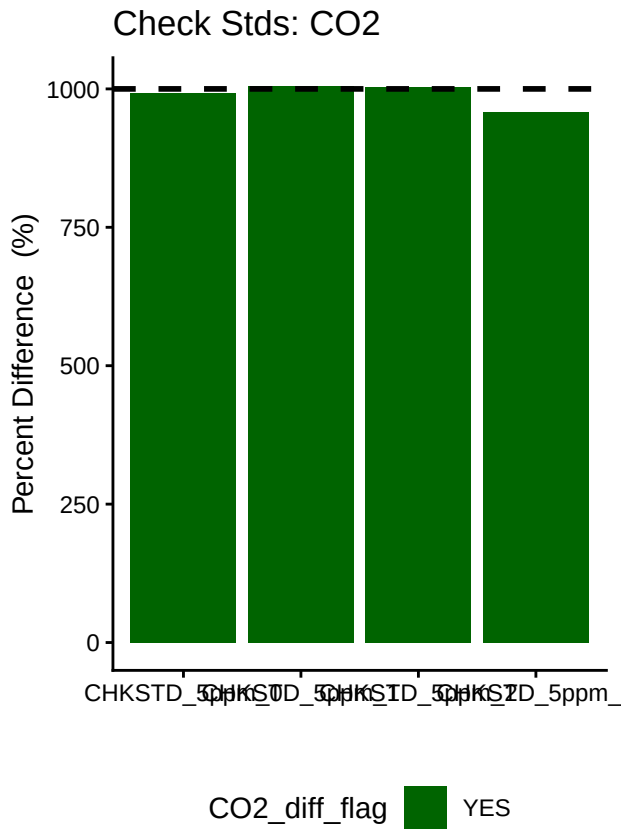


Now calculate the CH4 & CO2 concentrations in ppm

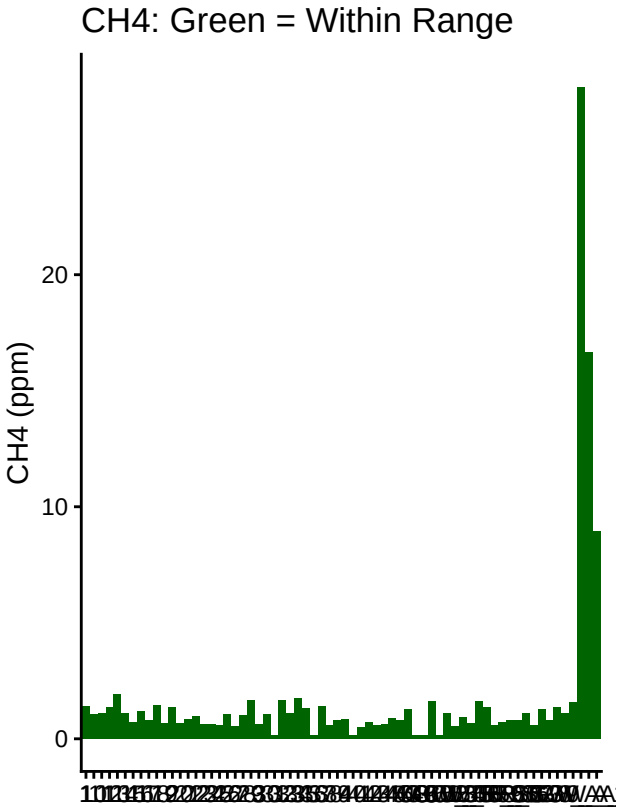
Check the Check Standards

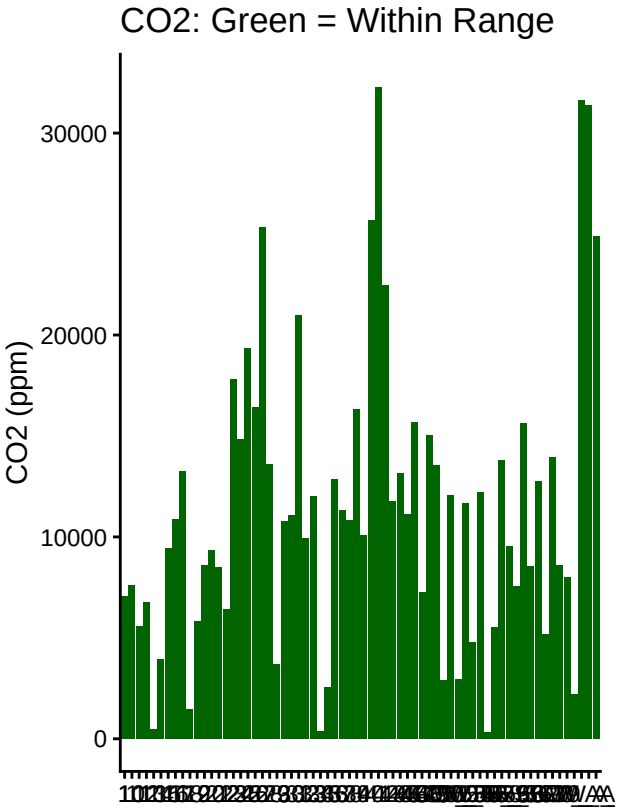
```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```


Check Stds: CH4
Percent Difference (%)
0
2
4
CHKSTD_5ppm_
CH4_diff_flag
YES

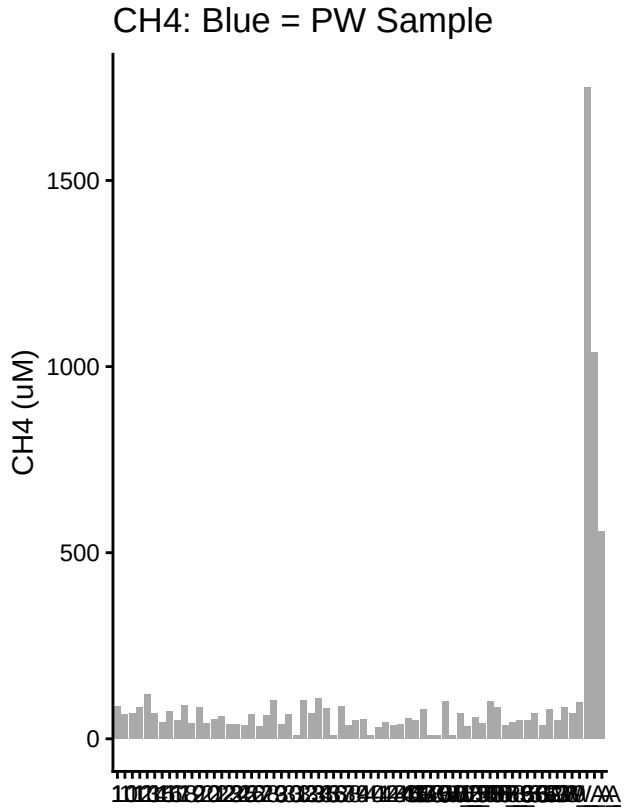

Check Stds: CO2
Percent Difference (%)
0
250
500
750
1000
CHKSTD_5ppm_
CO2_diff_flag
YES

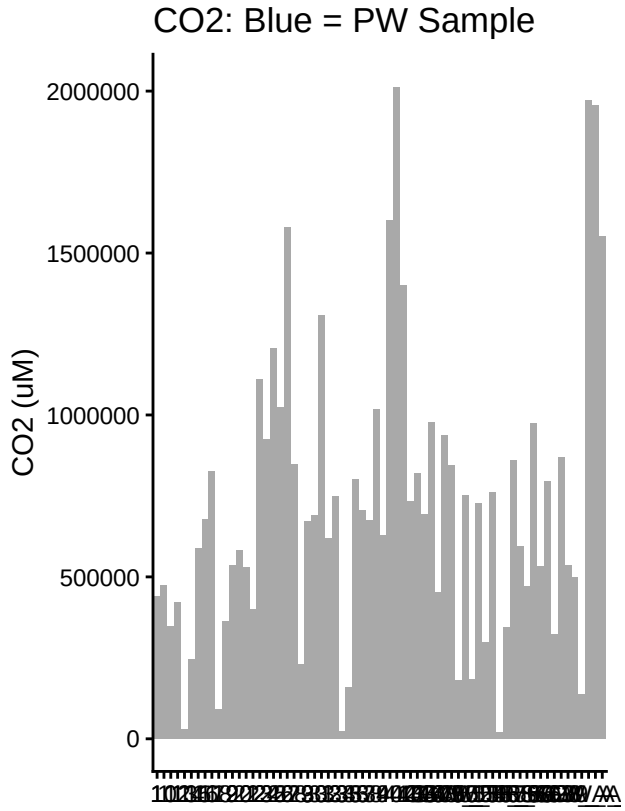
Dilution correct samples





If samples are water calculate gas in water





Write out processed data & slopes

```
#check results
#head(Samples)

#pull out what we need
Samples1 <- Samples[ ,c(1:3,6:16,20, 24:29)]
#head(Samples1)

write.csv(Samples1, "2023 Processed Data/COMPASS_TEMPEST_SGW_202304_Processed.csv")
```

#end